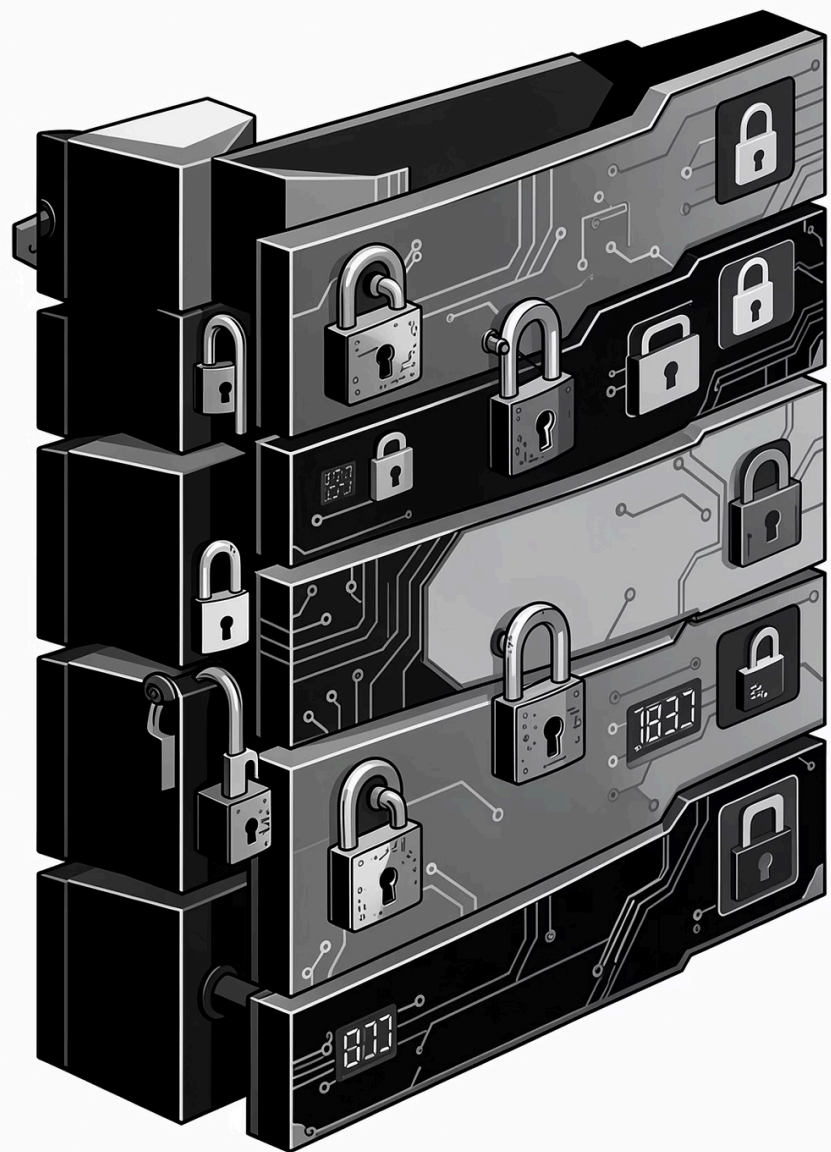




CyberShield: An AI-Powered Cyber Threat Detection System

A Final Year Project in Machine Learning and Network Security



The Security Challenge: Beyond Traditional Firewalls

Traditional Approach

Signature-based detection systems rely on known attack patterns stored in databases. They compare incoming traffic against predefined signatures.

Critical Limitations

- Cannot detect zero-day attacks without prior signatures
- High false negative rates on novel threats
- Requires constant database updates
- Ineffective against polymorphic malware

Intelligent Detection: The CyberShield Solution



Behavioral Analysis

Machine Learning models analyze network behavior patterns rather than static signatures



20 Network Features

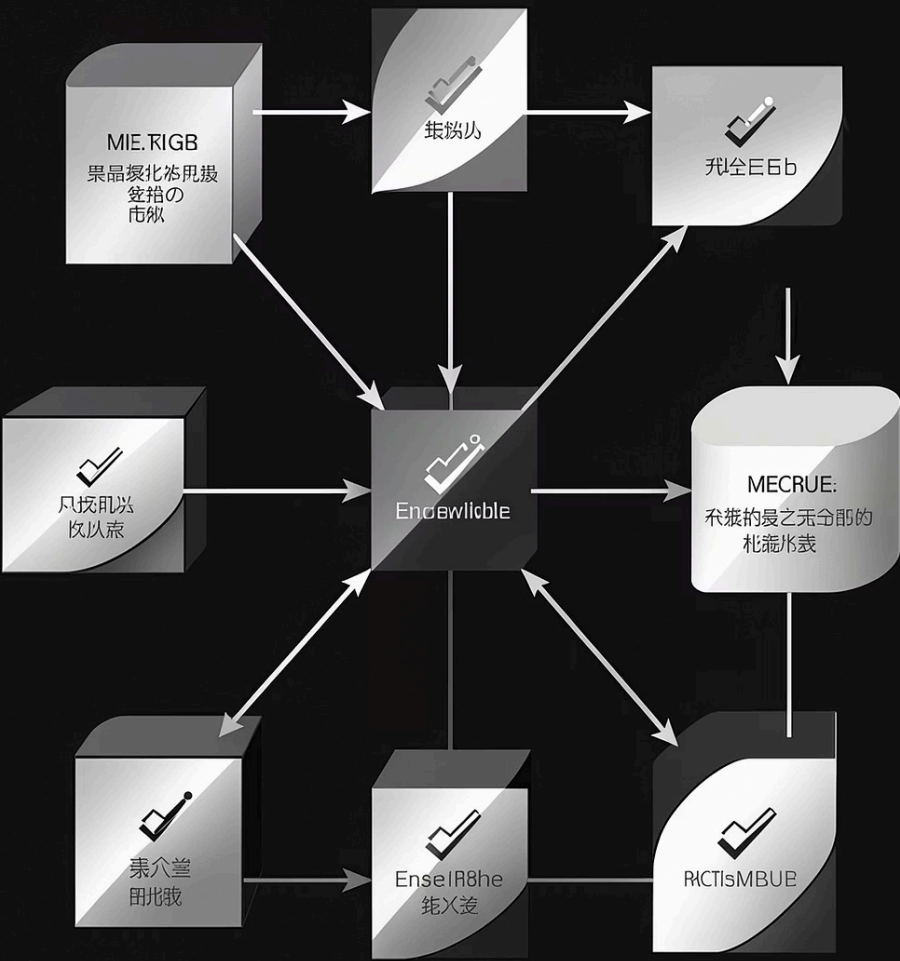
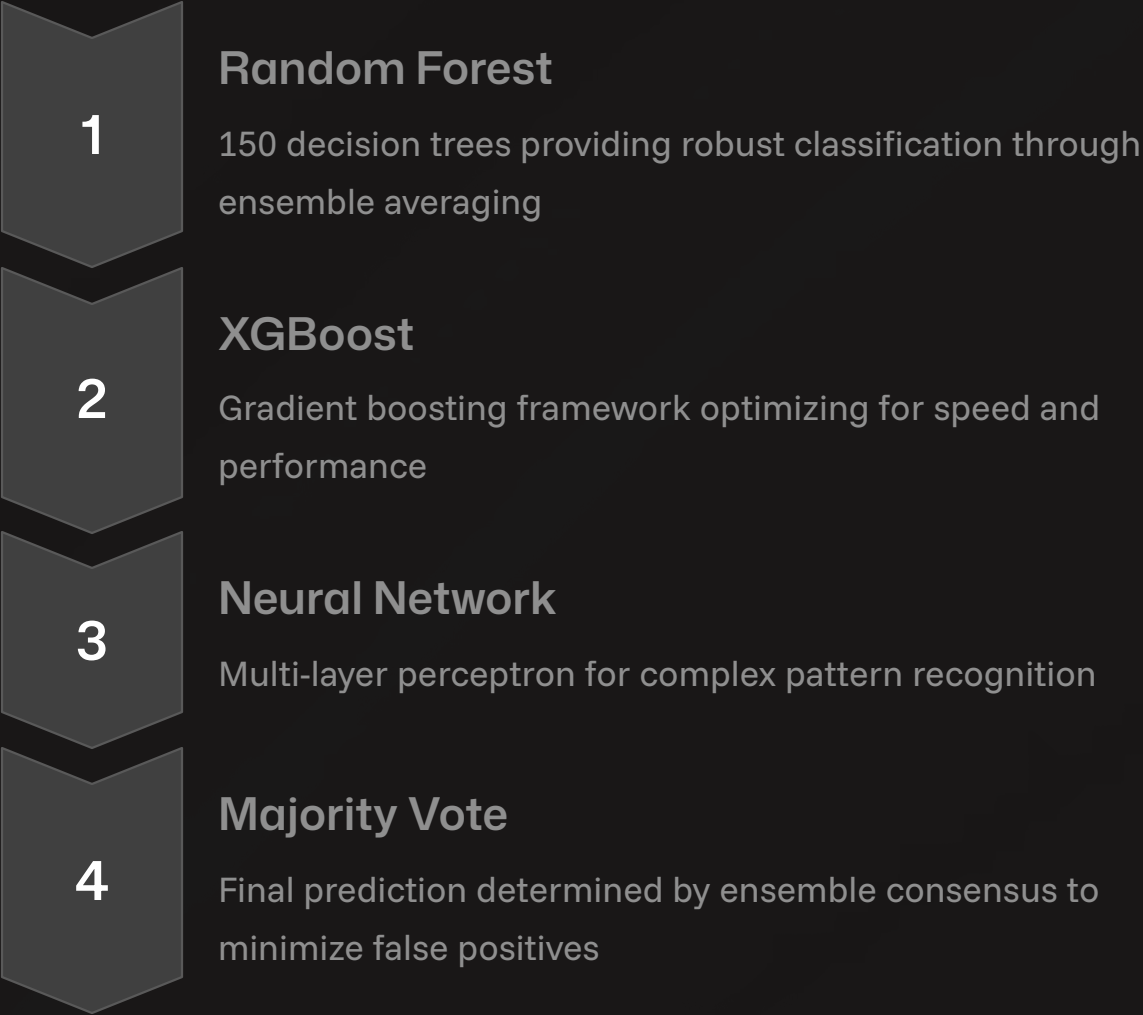
Comprehensive feature extraction including failed logins, error rates, connection duration, and traffic patterns



7 Threat Types

Simultaneous detection of DDoS, Botnet, Brute Force, SQLi, XSS, Port Scan, and Normal traffic

System Architecture: Ensemble Learning Design



Technology Stack and Implementation



Python Core

Backend implementation using Python 3.x with object-oriented architecture



Flask REST API

Lightweight web framework handling HTTP requests and model inference



ML Libraries

Scikit-learn for Random Forest, XGBoost for gradient boosting, NumPy for numerical computing



Live Dashboard

Custom HTML/JS/CSS frontend with cyberpunk aesthetic displaying real-time threat analysis

Ensemble Model: Performance Characteristics

Random Forest Configuration

150 decision trees with bootstrap aggregation and feature randomness to prevent overfitting. Each tree votes independently on threat classification.

XGBoost Optimization

Sequential model building with gradient descent optimization. Handles imbalanced datasets effectively through weighted class distributions.

Neural Network Architecture

Multi-layer perceptron with input layer (20 features), hidden layers with ReLU activation, and softmax output for probability distribution across 7 classes.

Majority Voting Logic

Each packet analyzed by all three models simultaneously. Final classification requires at least 2/3 models to agree, significantly reducing false positives while maintaining sensitivity.



Live Dashboard and Data Simulation

Real-Time Visualization

Dynamic dashboard displays incoming network packets, threat classifications, confidence scores, and historical attack patterns with color-coded severity indicators.

Data Simulator

Generates realistic synthetic network traffic with configurable parameters. Produces authentic feature distributions mimicking actual attack vectors for demonstration purposes.



Critical Clarification: The data simulator generates input features, but the AI models are fully operational and mathematically classify each packet in real-time using trained machine learning algorithms.

Packet Analyzer: Manual Testing Interface

01

Feature Injection

02

Model Processing

03

Consensus Decision

Majority vote determines final classification displayed on screen in real-time

04

Validation Proof

Mathematical demonstration shows AI switching prediction to 'Brute Force' when failed login threshold exceeded



Technical Challenges and Solutions

Model Integration Complexity

Challenge: Synchronizing three distinct ML algorithms with different input requirements

Solution: Built unified preprocessing pipeline normalizing all features before distribution to individual models

Real-Time Performance

Challenge: Processing packets at network speed without latency

Solution: Implemented asynchronous processing with thread pooling and model caching for rapid inference

False Positive Reduction

Challenge: Balancing sensitivity with specificity in threat detection

Solution: Majority voting ensemble architecture combined with confidence thresholding eliminates low-confidence classifications



Conclusion and Future Enhancements

Project Achievements

- Successfully implemented ensemble ML architecture
- Achieved robust zero-day attack detection capability
- Built complete REST API with live dashboard
- Created comprehensive testing framework

Future Roadmap

- Integration with real network hardware
- Adaptive learning from new attack patterns
- Cloud deployment for distributed monitoring
- Automated incident response capabilities

CyberShield represents a significant advancement in behavioral threat detection, demonstrating practical application of ensemble machine learning to real-world security challenges.